Empirical Transition Prior Table Successfully Computed

I've successfully executed the SQL query to compute the **Empirical Transition Prior table** with Laplace smoothing based on second-order Markov dependencies (lag-1 and lag-2 payouts).

Query Overview

The query performs the following operations:

1. **Self-joins history table** to create lag-1 and lag-2 payout sequences for consecutive ticks
2. **Joins with queries table** to obtain machine_type and signal metadata
3. **Groups by** machine_type, signal, lag1_payout, and lag2_payout
4. **Computes statistics** for each state combination:
• Raw counts for each payout outcome (0, 1, 2, 5, 20)• Laplace-smoothed probabilities: p_x = (count_x + 0.5) / (sample_count + 2.5)• Expected payout: E[payout] = 1×p_1 + 2×p_2 + 5×p_5 + 20×p_20
5. **Filters** for sample_count ≥ 10 to ensure statistical reliability

Results Summary

The complete transition table contains **194 rows** representing different state combinations across:

• **3 machine types**: balanced, jackpot, steady
• **3 signal values**: -1, 0, 1
• **5 lag1_payout values**: 0, 1, 2, 5, 20
• **5 lag2_payout values**: 0, 1, 2, 5, 20

Key Insights from the Data

Machine Type Characteristics:

• **Jackpot machines** show high p_0 probabilities (often 0.6-0.9), indicating frequent zero payouts with occasional large rewards
• **Balanced machines** have more uniform distributions across payout values
• **Steady machines** show moderate p_0 values (0.3-0.5) with higher probabilities for small payouts (1, 2)

Signal Effects:

• Signal values (-1, 0, 1) modulate the transition probabilities within each machine type
• Positive signals tend to increase expected payouts
• The signal effect varies by machine type and previous payout history

Sample Sizes:

• Range from 10 (minimum threshold) to 8,988 observations
• Larger sample sizes occur for common state transitions (e.g., consecutive zeros on jackpot machines)
• Rare transitions (e.g., 20→20) have smaller sample sizes but are still included when n≥10

Example Interpretation

For a **balanced machine** with **signal=0**, after payouts of **lag2=0, lag1=0**:

• Sample count: 6,378 observations

• p_0 = 0.676 (67.6% chance of payout 0)
• p_1 = 0.118 (11.8% chance of payout 1)
• p_2 = 0.143 (14.3% chance of payout 2)
• p_5 = 0.053 (5.3% chance of payout 5)
• p_20 = 0.010 (1.0% chance of payout 20)
• **Expected payout = 0.868**

This transition table can be used for:

• **Bayesian inference** with the Laplace-smoothed priors
• **Predictive modeling** of next-tick payouts
• **Policy optimization** in reinforcement learning contexts
• **Risk assessment** and variance estimation